

Inverse Suitability: Identifying Condition Difficulty and Rider Skill from Behavioural Outcomes via Continuous-Item Response Theory

Fabio Carucci*

Goable

<https://goable.io>

July 1, 2026

Abstract

Suitability scoring for outdoor activities—kitesurfing, paragliding, ski touring—maps environmental conditions to a go/no-go verdict via expert-defined curves. These curves conflate two distinct quantities: the intrinsic difficulty of a condition and the skill of the person facing it. We introduce *Inverse Suitability*, a continuous-item Item Response Theory (IRT) model that identifies both from behavioural outcomes alone. Each outcome is a triple (rider r , condition metric x at site s , binary outcome y); we model $P(y=1) = \sigma(a(\theta_r - \delta(x, s)))$, where θ_r is latent rider skill, $\delta(x, s)$ is a latent difficulty function anchored to a physics-derived expert curve as its prior, and a is a discrimination parameter. The formulation is strictly more general than a single suitability curve, which it recovers exactly when skill is integrated out under the population distribution. Parameters are estimated by marginal maximum likelihood with Gauss–Hermite quadrature; identification holds when the rider $\times$ condition incidence graph is connected, with a documented single-curve fallback otherwise. We validate via synthetic recovery: on a reference cohort (80 riders $\times$ 30 outcomes) the model recovers latent skill at $r = 0.96$, locates the difficulty minimum within ± 3 units of ground truth, and improves held-out Brier Skill Score by +0.33 over the expert-curve baseline. The recovered difficulty function defines a measurable, site-level construct—an intrinsic difficulty atlas—that existing meteorological observation networks do not capture. All results reproduce from a single command on synthetic data, requiring no proprietary observations.

Keywords: item response theory; latent-variable models; suitability scoring; marginal maximum likelihood; environmental decision science; skill identification.

1 Introduction

Operational decision support for outdoor activities rests on *suitability scoring*: a mapping from the environmental state—wind speed, wave height, thermal lift, snow stability—to a bounded score or a go/no-go verdict. In practice these mappings are expert-defined curves, one per activity and

*The method is developed and implemented within *Goable*’s environmental calibration engine. The reproducibility artifact is the open command `inverse-suitability latent-demo`, released with the paper source at <https://github.com/goable-io/inverse-suitability-paper> (Section 5.3). Correspondence: research@goable.io.

per physical dimension, calibrated by domain specialists and, increasingly, refined against observed outcomes. A kitesurfing wind curve peaks near a sweet spot and falls off toward calm and storm; a ski-touring stability curve gates on avalanche risk. The score is then a composition of such curves.

A single suitability curve, however, silently conflates two quantities that a decision-maker needs to keep apart. The first is the *intrinsic difficulty* of a condition: how objectively demanding a given wind or sea state is for the activity, independent of who is attempting it. The second is the *skill* of the person facing that condition. When a curve is fit to behavioural outcomes—did the session go well, did the rider go out at all—it absorbs both: the apparent “suitability” of 22 knots at a given spot reflects not only how hard 22 knots is, but also the skill distribution of whoever happens to ride there. Two spots with identical fitted suitability curves can therefore have radically different intrinsic difficulty, masked by different clienteles. No existing observation network—marine, atmospheric, cryospheric—measures the difficulty term in isolation, because none carries the human-decision layer needed to identify it.

The gap. What is missing is a way to *separate* intrinsic condition difficulty from population skill using the outcome data that decision platforms already collect, and to do so with an identified, interpretable model rather than a black box. The methodological obstacle is that conditions are not discrete test items: wind speed is a continuous axis, not a yes/no question, so the classical psychometric machinery for separating item difficulty from person ability does not apply off the shelf.

Contributions. This paper makes three contributions.

1. **Continuous-item IRT for environmental conditions.** We extend explanatory item response theory—which already lets item difficulty depend on covariates [Fischer, 1973, De Boeck and Wilson, 2004]—from discrete item features to a *continuous physical driver*, modelling difficulty $\delta(x, s)$ as a smooth, physically anchored function of a metric x at site s (Section 3). The item bank is the weather.
2. **Identifiability and synthetic recovery.** We give the conditions under which rider skill and condition difficulty separate—a connected rider $\times$ condition incidence graph plus location/scale anchoring—and a deterministic marginal-maximum-likelihood estimator with a documented single-curve fallback when identification fails (Section 4). We validate by synthetic recovery: on a reference cohort the model recovers latent skill at correlation 0.96 and locates the difficulty minimum within ± 3 units of ground truth (Section 5).
3. **The difficulty atlas.** The recovered difficulty function defines a measurable, site-level construct—an *intrinsic difficulty atlas*—that is anonymous, joinable to existing meteorological archives, and captures a quantity those archives do not (Section 6).

Scope. This is a methodology paper. Its validation is a synthetic recovery study: cohorts are generated with known ground-truth skill and difficulty, and the estimator is shown to recover them. We make *no* empirical claim on real riders; no proprietary observations are used anywhere, and every number in the paper reproduces from a single command on synthetic data (Section 5). Empirical validation on field cohorts is future work (Section 8).

2 Related work

Our contribution sits at the intersection of three literatures that have not previously been connected: psychometric latent-trait models, meteorological suitability indices, and latent-variable modelling in the environmental sciences.

Item Response Theory. IRT is the standard framework in psychometrics and educational measurement for separating the difficulty of a test item from the ability of a test-taker, using only binary (correct/incorrect) responses [Lord and Novick, 1968, Rasch, 1960, Embretson and Reise, 2000]. The two-parameter logistic model writes the probability of a correct response as $P(y=1) = \sigma(a_i(\theta_p - b_i))$ for person ability θ_p , item difficulty b_i , and item discrimination a_i [Birnbaum, 1968]. Parameters are classically estimated by marginal maximum likelihood, integrating over the person-ability distribution with an EM algorithm and Gauss–Hermite quadrature [Bock and Aitkin, 1981]; Baker and Kim [2004] survey the estimation techniques. The classical model treats items as *discrete*, each carrying its own scalar difficulty b_i estimated from the responses to that item. Two established lines of work already relax this by letting difficulty depend on covariates: the linear logistic test model (LLTM), which writes item difficulty as a *linear* combination of item features [Fischer, 1973], and the broader family of *explanatory* item response models, which embed item and person effects in a generalized linear (or nonlinear) mixed model [De Boeck and Wilson, 2004]. Our model is a continuous-item member of this explanatory family. It extends LLTM/explanatory IRT along four axes: (i) the difficulty covariate is a *continuous physical driver* (e.g. wind speed), not a set of discrete item features; (ii) difficulty is a *smooth* function (spline or Gaussian process) rather than LLTM’s linear form; (iii) it is *anchored to a physics-derived curve* so its units are physically interpretable; and (iv) it is applied to a new domain—environmental suitability—in which no two sessions share an identical item, so the discrete-item view is not merely inconvenient but inapplicable. Positioned this way, the contribution is an explicit extension of explanatory IRT to a continuous environmental driver, not a claim that difficulty has never been modelled as a function.

Suitability and climate indices. Environmental decision science has a long tradition of expert-defined indices that map weather to a rating. The Tourism Climate Index [Mieczkowski, 1985] combines thermal comfort, precipitation, sunshine, and wind into a single score; the Holiday Climate Index [Scott et al., 2016] refines the weighting scheme, and second-generation indices introduce climatic thresholds [de Freitas et al., 2008]. Activity-specific suitability curves used in operational scoring engines are of the same family: expert-weighted transformations of physical variables. These indices are, by construction, population- and expert-level: they express how suitable a condition is *on average*, and none decomposes that rating into an intrinsic-difficulty term and a skill term. Inverse Suitability takes such an expert curve as the *prior* for its difficulty function and then identifies the residual structure that the single curve cannot express.

Latent-variable models in environmental science. Latent-variable and random-effects models are widely used in ecology and environmental statistics—joint species distribution models built on latent factors [Warton et al., 2015], occupancy models that separate a latent “true state” from an imperfect detection process [MacKenzie et al., 2002], and item-factor-analytic treatments of ordinal environmental data. The closest in spirit are occupancy models: our skill term plays a role analogous to a site- or observer-specific detection propensity, separating a latent property of interest from the process that reveals it. What is new relative to these is (i) anchoring the latent difficulty to a physics-derived curve so that it is measured in interpretable units rather than recovered only up

to an arbitrary scale, and (ii) treating the difficulty as a smooth function of a continuous physical driver rather than a per-site constant.

Forecast verification. Our promotion metric, the Brier Skill Score, is standard in forecast verification [Brier, 1950]: it measures the mean-squared-error improvement of a probabilistic prediction over a reference. We use it to test whether skill-conditioning earns its added complexity against the single-curve baseline (Section 5).

3 The model

3.1 Outcomes and the link function

The unit of data is an outcome triple (r, x, y) , optionally indexed by a site s : rider r experienced a condition summarised by the metric value x (for example, wind speed) at site s and produced a binary outcome $y \in \{0, 1\}$ —a good session versus not, or a go versus no-go. We model the outcome probability with a two-parameter latent factorisation,

$$P(y=1 \mid r, x, s) = \sigma\left(a(x, s) (\theta_r - \delta(x, s))\right), \quad \sigma(z) = \frac{1}{1 + e^{-z}}, \quad (1)$$

with three latent objects:

- $\theta_r \in \mathbb{R}$ — the *latent skill* of rider r , one scalar per rider (per pseudonym), identified up to a location/scale convention (Section 4);
- $\delta(x, s)$ — the *latent difficulty* of the condition: how objectively hard the metric value x is at site s , a smooth function of x ;
- $a(x, s) > 0$ — the *discrimination*: how sharply the condition separates riders of different skill. In the Phase-1 form used throughout this paper, a is a single scalar.

Equation (1) is the two-parameter logistic IRT model with the item difficulty b_i replaced by a function $\delta(x, s)$ of a continuous driver. Higher skill or lower difficulty raises the success probability; the two enter only through their difference $\theta_r - \delta(x, s)$, a fact with consequences for identification (Section 4).

3.2 Difficulty as a continuous item

Classical IRT difficulty is a per-item scalar because test items are discrete. Environmental conditions are not: the metric x is a continuum, so we model $\delta(x, s)$ as a smooth function—a spline (or, with full Bayesian inference, a Gaussian process) over x —rather than a collection of independent per-condition constants. Two properties make this well-posed. First, smoothness lets neighbouring conditions share statistical strength, so a condition observed only a few times still receives a sensible difficulty. Second, the difficulty function is *anchored to the physics-derived expert curve as its prior*, which serves two distinct purposes: it fixes the difficulty scale in interpretable, physical units, and it acts as a shrinkage prior toward which sparsely observed conditions regress. This anchoring is about *units and regularisation*, not identification: δ is already identified once the skill distribution is standardised (Section 4); without the anchor it would still be identified, but expressed only on an arbitrary scale.

3.3 Recovering the single suitability curve as a marginal

The model in (1) is strictly more general than today’s single suitability curve, and reduces to it exactly. Let $p(\theta)$ be the population skill distribution. The single suitability curve is the marginal of (1) over that population,

$$\text{curve}(x) = \mathbb{E}_{\theta} [\sigma(a(x)(\theta - \delta(x)))] = \int \sigma(a(x)(\theta - \delta(x))) p(\theta) d\theta. \quad (2)$$

Integrating skill out recovers precisely the population-level curve that a conventional calibrator would fit. This is not merely an asymptotic correspondence; in our implementation it is a regression test—the fitted model’s marginal is checked against a single-curve fit on the same data (Section 5, the *marginal matches single curve* gate). Consequently, adopting Inverse Suitability adds a dimension to an existing scoring engine; it does not fork it. Where a cell lacks the data to identify the latent structure, the model falls back to (2), which is the behaviour the engine already has.

3.4 Skill-conditioned scoring

For a chosen skill level θ —for instance a population quantile mapped to beginner, intermediate, or expert—the skill-conditioned suitability curve is $x \mapsto \sigma(a(x)(\theta - \delta(x)))$. This is the per-level scoring surface: the same condition is scored differently for a beginner and an expert, while the underlying difficulty δ is invariant (Figure 1). The skill level is treated as an explicit input, not inferred from identity—a boundary we return to in Section 7.

4 Identification and estimation

Separating skill from difficulty is the make-or-break of the method. Three obstacles stand in the way, each with an explicit remedy; the estimator is otherwise the classical IRT one.

4.1 Scale indeterminacy

Because (1) depends on skill and difficulty only through the difference $\theta_r - \delta(x, s)$, adding a constant c to every skill and to every difficulty leaves all outcome probabilities unchanged. Skill and difficulty are therefore identified only up to a common shift (and, with a free discrimination, a common scale). We use two devices with *distinct roles*—one that secures identification, one that fixes interpretable units:

1. **Location/scale on skill (identification).** We fix the population skill distribution to $\mathbb{E}[\theta] = 0$ and $\text{Var}[\theta] = 1$ —the standard IRT convention. This removes the shift-and-scale freedom and is what makes θ and δ separately identified, not merely their difference.
2. **Physical anchor on difficulty (units and prior).** Given identification, δ is still expressed on an arbitrary scale. Anchoring it to the physics-derived expert curve through its prior fixes the difficulty origin and units to interpretable, physical quantities, and simultaneously acts as a shrinkage prior for sparsely observed conditions (Section 4.3). It is not what identifies the model.

4.2 The data condition: a connected incidence graph

Identification also requires the right *coverage*. Form the bipartite incidence graph whose two vertex classes are riders and condition bins, with an edge whenever a rider is observed in a bin. Skill and difficulty separate only when this graph is *connected*: riders must be observed across varied conditions, and conditions must be experienced by varied riders. The failure mode is intuitive—a rider who only ever goes out in easy conditions has a skill that is confounded with “always succeeds,” because nothing in the data forces the two apart. Before fitting, the estimator checks

- connectivity of the rider $\times$ condition incidence graph;
- at least K distinct condition bins per rider;
- at least M riders per condition bin.

When these fail, the latent model *auto-disables* and the estimator falls back to the single-curve marginal of Section 3.3. This is a deliberate refusal to over-claim: below the identification threshold, the honest output is the population curve, not a spuriously “personalised” one. It also makes the activation gate strictly stronger than a raw sample-size threshold—volume alone does not identify skill; repeated cross-condition observation does.

4.3 Separation and sparsity

Two finite-sample pathologies remain. A rider with an all-success record drives $\theta \rightarrow +\infty$ (the classical separation problem in logistic models); a condition bin observed too rarely has an under-determined difficulty. Both are handled by shrinkage: rare skills shrink toward the population prior, and sparse difficulties shrink toward the expert-curve anchor. This keeps every estimate finite and well-defined without introducing new machinery beyond the priors already stated.

4.4 Estimator: marginal maximum likelihood

We estimate by *marginal maximum likelihood* via EM, the classical IRT estimator [Bock and Aitkin, 1981]. The skill of each rider is a latent variable; the marginal likelihood of the observed outcomes integrates it out against the population distribution $p(\theta) = \mathcal{N}(0, 1)$. The E-step evaluates this integral by Gauss–Hermite quadrature [Golub and Welsch, 1969]; the M-step maximises the difficulty function δ and the discrimination a given the current quadrature weights. The procedure is deterministic—same data and same seed give the same result—which matches the reproducibility posture of Section 7 and adds no heavy sampling dependency.

Full Bayesian estimation (for example Hamiltonian Monte Carlo with a Gaussian process difficulty) yields richer posterior uncertainty bands and is a documented upgrade; it is not required for the point-identification and recovery results of this paper, and we treat it as future work (Section 8). The per-rider skill posteriors and the difficulty function that the estimator returns are exactly the objects validated next.

5 Synthetic recovery experiment

Because no real cohort is used, the validity of the method rests on a *synthetic recovery* study: we generate data from a known ground truth, fit the model, and measure how well it recovers that truth. This is the standard validation for a methods paper, and it is the same role a simulation study plays throughout statistics—it isolates the estimator’s behaviour from the confounds of any particular dataset.

Table 1: Synthetic recovery on the reference cohort (80 riders $\times$ 30 paired outcomes, $n_{\text{train}} = 2,400$). Every value is recovered from behavioural outcomes alone by marginal maximum likelihood; none is supplied to the fit. The data-generating difficulty is minimised at the ground-truth optimum $x^* = 18$ kn with true discrimination $a_{\text{true}} = 2.0$ (stated in the text, not fit here). Reproduced by a single command (`inverse-suitability latent-demo`); see Appendix.

Quantity	Symbol	Recovered
Latent-skill recovery (Pearson)	$\text{corr}(\theta, \hat{\theta})$	0.964
Difficulty-minimum location	$\hat{x}^*$ (kn)	16.7
Discrimination	$\hat{a}$	2.16
Held-out Brier skill score	BSS vs. single curve	+0.329
Marginal matches single curve	–	✓
Identification gate passed	–	✓

5.1 Data-generating process

The generator draws a two-factor cohort with *known* skill and difficulty. Ground-truth skills are $\theta_r \sim \mathcal{N}(0, 1)$, one per rider. The ground-truth difficulty is a parabola in difficulty space,

$$\delta(x) = \left(\frac{x - x^*}{w} \right)^2 + f, \quad x^* = 18 \text{ kn}, w = 8, f = -1, \quad (3)$$

so that difficulty is *minimised at the ground-truth optimum* $x^* = 18$ knots and rises symmetrically toward calm and storm—a single-peaked suitability profile once passed through the link. Each rider is observed across conditions drawn uniformly over $[3, 35]$ knots (guaranteeing a connected incidence graph, Section 4.2), and outcomes are Bernoulli draws from (1) with *true discrimination* $a_{\text{true}} = 2.0$. The reference cohort has 80 riders $\times$ 30 outcomes each. These parameters— $x^* = 18$ kn, $a_{\text{true}} = 2.0$, and the cohort shape—are the ground truth supplied to the *generator*; they are never supplied to the *fit*. The recovered quantities reported below are what the estimator infers from the outcomes alone.

5.2 Metrics and results

We assess recovery on three axes: (i) latent-skill recovery, the Pearson correlation between the true θ_r and the estimated $\hat{\theta}_r$; (ii) difficulty-shape recovery, summarised by the estimated location of the difficulty minimum $\hat{x}^* = \arg \min_x \hat{\delta}(x)$, which should land near the true 18 kn; and (iii) predictive gain, the held-out Brier Skill Score of the skill-conditioned model against the single-curve baseline under a stratified 80/20 split. We also verify the backward-compatible identity of Section 3.3—that integrating skill out of the fitted model matches a single-curve fit—and that the identification gate of Section 4.2 passes on this (deliberately connected) cohort.

Table 1 reports the outcome. The model recovers latent skill at correlation 0.96 against ground truth; it locates the difficulty minimum at $\hat{x}^* = 16.7$ knots, within ± 3 knots of the true 18-knot optimum; and it improves the held-out Brier Skill Score by +0.33 over the single-curve baseline, confirming that skill-conditioning earns its added complexity out of sample rather than merely fitting the training data more flexibly. The recovered discrimination is $\hat{a} = 2.16$ (true $a_{\text{true}} = 2.0$), the marginal matches the single-curve fit, and the identification gate passes, all on $n_{\text{train}} = 2,400$ training outcomes.

5.3 Reproducibility

The experiment is a single command. Every value in Table 1 is emitted by

```
uv run inverse-suitability latent-demo --out DIR
```

which fits the reference cohort and writes a `stats.json`; the paper’s table and inline figures are rendered directly from that file, so the reported numbers cannot drift from the code (Appendix B). The run is deterministic (fixed cohort and fit seeds), completes in well under a minute on a laptop, touches no proprietary data, and requires no network access. The data-generating process is fully specified by (3) and Section 5.1; a reviewer can regenerate the cohort and re-fit independently.

6 The difficulty atlas

The estimator returns two kinds of object. One is per-rider: the skill posteriors $\hat{\theta}_r$. The other is a property of the environment: the difficulty function $\hat{\delta}(x, s)$ itself. Collected across sites and dimensions, the latter forms what we call the *intrinsic difficulty atlas*—a site-level, anonymous map of how hard each condition is, net of who rides it.

6.1 Why the atlas is new

Existing observation networks—marine reanalyses, atmospheric mesonets, lightning-detection and air-quality networks, national avalanche services—densely cover the *meteorological* state. What none of them holds is the quantity δ isolates: the intrinsic difficulty of a condition for a skilled human activity, separated from the population skill mix that otherwise confounds it. Suitability, as conventionally fit, is a blend of difficulty and clientele; δ is the difficulty term alone. Because it is anchored to a physical curve (Section 3), it is expressed in interpretable units and is *joinable* to those same meteorological archives: for any site and condition, the atlas adds a difficulty coordinate the archives lack.

6.2 A measurable construct

The consequence is conceptual as much as practical. Two spots with identical fitted suitability curves can have very different intrinsic difficulty, masked by different clienteles; until the skill term is identified, they are indistinguishable. Inverse Suitability makes them distinguishable, and in doing so treats activity suitability—for the first time, to our knowledge—as a formal, measurable latent construct rather than an expert convention. The atlas is the artifact of that construct: a difficulty rating decoupled from who happens to be present, defined precisely enough to be estimated, reproduced, and audited.

7 Governance: reproducibility and privacy

A latent model estimated from human behavioural data must be reproducible enough to audit and private enough to respect the people in it. The two requirements pull in opposite directions—auditability wants to retain data, privacy wants to delete it—and the design resolves them by separating what is anonymous from what is personal.

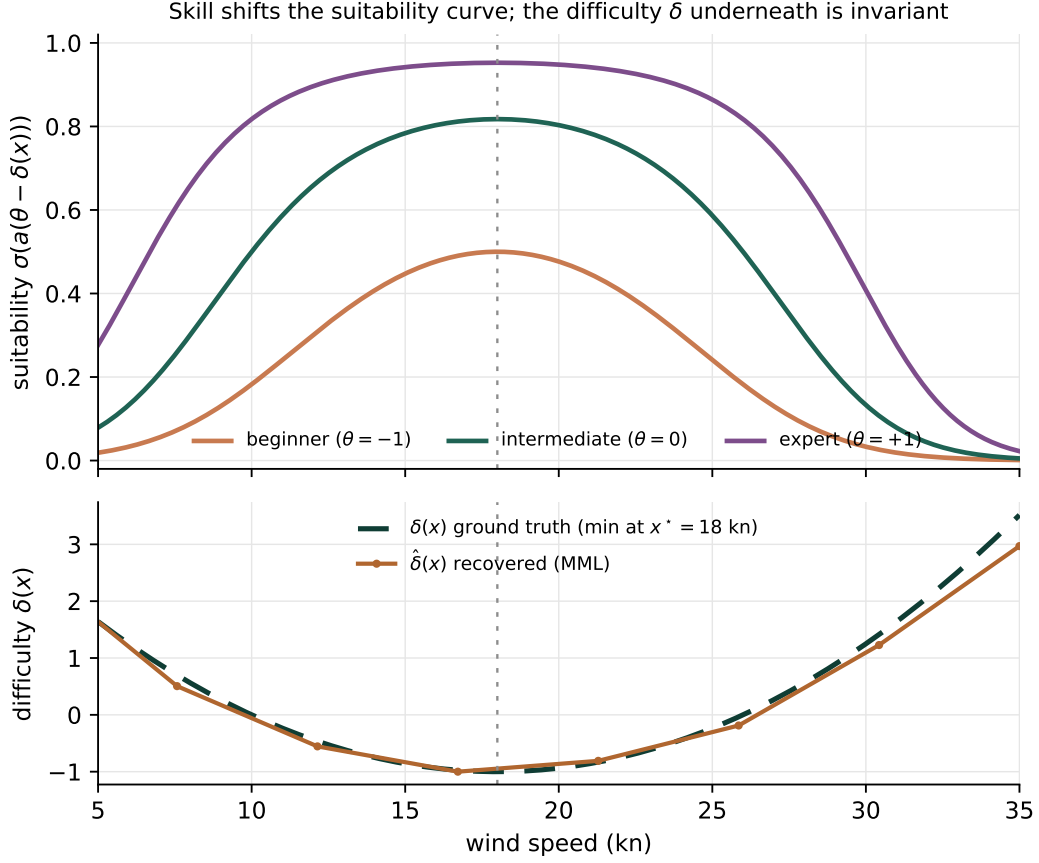


Figure 1: The construct the atlas measures. *Bottom*: the intrinsic difficulty function $\delta(x)$ (dashed), minimised at the ground-truth optimum $x^* = 18$ kn; the recovered $\hat{\delta}(x)$ is overlaid where available. *Top*: three skill-conditioned suitability curves $\sigma(a(\theta - \delta(x)))$ for beginner ($\theta = -1$), intermediate ($\theta = 0$), and expert ($\theta = +1$). Skill shifts the suitability curve while the difficulty beneath it is invariant: two operators whose spots share the same δ but differ in clientele read different suitability surfaces from the *same* underlying difficulty. That separation is what no meteorological observation network records. Figure generated from code (Appendix B).

7.1 Reproducible by construction

Each fit is pinned to a content hash of the exact cohort it consumed (a SHA-256 of the canonicalised outcome table). A cited estimate—a difficulty function or a rider’s skill posterior from cohort hash `abc123...`—can be re-derived by re-running the estimator on the cohort that hash references, without the authors’ cooperation. Combined with the deterministic marginal-maximum-likelihood estimator (Section 4.4), this makes every published number independently checkable: same cohort hash and same seed yield the same posterior. Storing fits append-only, keyed by cohort hash, additionally turns a rider’s skill estimates over successive cohorts into a trajectory—a signal for whether skill estimates are stable or drift after a regime change.

7.2 Privacy: the anonymous–personal split

The two output classes have different privacy status, and the split is the crux.

- The **difficulty atlas** $\hat{\delta}(x, s)$ is a property of a *place*, not a person. It carries no per-rider information and is therefore anonymous; it survives deletion requests, in the same way that aggregate statistics do.
- The **skill posteriors** $\hat{\theta}_r$ are per-person attributes and are treated as sensitive: on a rider’s opt-out, every skill estimate for that rider is hard-deleted across all cohort versions. Deletion is a fan-out over the append-only history, not merely a removal of the latest value.

This separation is what lets the atlas remain a durable, publishable scientific artifact while the personal layer remains fully erasable. An anonymised public export of the atlas—generalised in space and lagged in time, released only for cells with enough distinct contributors—follows the same governance as other anonymised environmental exports and does not depend on retaining any individual’s data.

7.3 Skill is an input, not an inference

Finally, a boundary that is as much a governance choice as a modelling one: skill-conditioned scoring (Section 3) takes the skill level as an *explicit, caller-provided* input. The score for “expert” conditions is the expert curve; the system does not infer a person’s skill from their identity in order to score them. This keeps the objective, physics-anchored difficulty curve separate from any automatic profiling of individuals—skill-conditioned physics, not personalisation—and it means the difficulty atlas can be built and used without ever attaching a skill estimate to a named person.

8 Discussion and applications

Skill-conditioned decision support. The most direct application is scoring that adapts to a declared skill level. Because the difficulty term is shared and only the skill offset changes (Section 3), a single identified model yields a family of suitability surfaces—beginner through expert—at no additional fitting cost, each falling back invisibly to the population curve where the latent structure is not identified. The result is a verdict that answers “hard for whom,” not just “hard on average.”

An intrinsic-risk primitive for underwriting. Parametric insurance for weather-dependent leisure needs to price the intrinsic risk of a site, not the muddled suitability of whoever currently frequents it. The difficulty atlas is a natural credibility primitive for such pricing: a difficulty coordinate per site and condition, anchored to physics, reproducible from a cohort hash, and independent of the clientele that a suitability curve would otherwise bake in. We note the connection without building it here.

Limitations. The paper’s central limitation is deliberate and stated plainly: its validation is a *synthetic recovery study*, not an empirical evaluation on real riders. We show that when data are generated from the model, the estimator recovers the ground truth; we do *not* show that real behavioural outcomes follow the model, because no real cohort is analysed here. Several modelling simplifications accompany this scope. The discrimination a is a single scalar rather than a function of the condition; difficulty is modelled one physical dimension at a time rather than jointly over the full condition vector (wind×wave×cold), which would strengthen realism at the cost of harder identification; and estimation is point-based marginal maximum likelihood, so the reported skills and difficulties are point estimates rather than full posteriors. Finally, identification requires a connected rider×condition graph (Section 4.2); in real deployments many site×condition cells will not meet that bar and will—correctly—return the single-curve fallback rather than a latent fit.

Future work. Three extensions follow directly. First, *empirical validation*: fit the model to real field cohorts as they accrue, and test the model’s structural assumptions (for instance, whether the marginal identity of Section 3.3 holds on real data, and whether the held-out skill gain persists out of sample). Second, *full Bayesian inference* with a Gaussian-process difficulty, giving calibrated uncertainty bands on both the atlas and the skill posteriors. Third, *multi-dimensional difficulty*, jointly identifying δ over the condition vector, which turns the per-dimension atlas into a genuine multi-driver difficulty surface. Each is compatible with the identification and governance framework established here.

9 Conclusion

Suitability scores conflate two things a decision-maker must separate: how hard a condition is, and how skilled the person facing it is. We showed that a continuous-item Item Response Theory model— $P(y=1) = \sigma(a(\theta_r - \delta(x, s)))$ with a smooth, physics-anchored difficulty function— identifies both from binary behavioural outcomes alone. The model is strictly more general than a single suitability curve, which it recovers exactly on marginalising skill; it is identified when the rider×condition incidence graph is connected, and it fails safely to the single curve when it is not. A synthetic recovery study confirms the estimator recovers latent skill at correlation 0.96, locates the difficulty minimum within ± 3 units of ground truth, and improves held-out Brier Skill Score by +0.33 over the expert-curve baseline—all reproducible from one command on synthetic data, with no proprietary observations.

Beyond the estimator, the recovered difficulty function defines a measurable, anonymous, site-level construct—an intrinsic difficulty atlas—that existing meteorological networks do not capture and that is joinable to them. Establishing this construct, and the conditions under which it can be identified and governed, is the paper’s lasting contribution. What remains is empirical: to fit the model to real cohorts as they accumulate and test whether human behaviour in the field follows the structure that, in simulation, it recovers so cleanly.

Code and data availability

The L^AT_EX source of this paper and a self-contained reproducibility package—the `goable-calibration latent-demo` command together with the table- and figure-generation scripts—are released at <https://github.com/goable-io/inverse-suitability-paper> under the MIT licence (code) and CC BY 4.0 (text). The package reproduces every value in Table 1 and Figure 1 from synthetic data with a single command (`make repro`); no proprietary observations are required or included. The production calibration engine, the real behavioural-outcome cohorts, and any fitted difficulty atlases are *not* part of this release—this paper and its package concern the methodology and its synthetic validation only.

A Estimation details

Marginal maximum likelihood. Writing $\Phi = (\delta, a)$ for the structural parameters and $p(\theta) = \mathcal{N}(0, 1)$ for the population skill density, the marginal likelihood of rider r ’s outcomes $\{(x_{ri}, y_{ri})\}_i$ is

$$L_r(\Phi) = \int \prod_i \sigma(a(x_{ri})(\theta - \delta(x_{ri})))^{y_{ri}} (1 - \sigma(\cdot))^{1-y_{ri}} p(\theta) d\theta. \quad (4)$$

The EM algorithm alternates an E-step, which forms the posterior over each rider’s skill on a fixed quadrature grid, and an M-step, which maximises the expected complete-data log-likelihood over Φ . The full log-likelihood is $\sum_r \log L_r(\Phi)$.

Gauss–Hermite quadrature. The skill integral is evaluated by Gauss–Hermite quadrature [Golub and Welsch, 1969]: nodes $\{z_q\}$ and weights $\{w_q\}$ approximate

$$\int f(\theta) \mathcal{N}(\theta; 0, 1) d\theta \approx \sum_q w_q f(z_q)$$

after the standard change of variables. The node count trades accuracy against cost; the demonstration configuration uses a fixed grid adequate for a unimodal skill posterior.

Difficulty representation and anchoring. The difficulty function is represented on a metric grid and evaluated by interpolation. Its prior is centred on the physics-derived expert curve, which anchors the difficulty origin and scale (Section 4); rare or unobserved conditions shrink toward that anchor. Rider skills carry the $\mathcal{N}(0, 1)$ population prior, which both sets the location/scale convention and shrinks separated (all-success or all-failure) riders to finite values.

Determinism. Given a cohort and a seed, the estimator is deterministic—no stochastic sampling enters the point estimate—so results are bit-for-bit reproducible.

B Data-generating process and reproducibility

Generator. The reference cohort is drawn as in Section 5.1: 80 riders, 30 outcomes each; skills $\theta_r \sim \mathcal{N}(0, 1)$; conditions uniform on $[3, 35]$ knots; ground-truth difficulty (3) with optimum $x^* = 18$ kn, width $w = 8$, floor $f = -1$; true discrimination $a_{\text{true}} = 2.0$; cohort seed fixed. Outcomes are Bernoulli draws from the model (1). These are the ground-truth inputs to the *generator* and are not provided to the fit.

One-command reproduction. The results in Table 1 are produced by

```
uv run inverse-suitability latent-demo --out DIR
```

which fits the cohort and writes `stats.json`. Under `reproducibility/`, the paper’s results table and inline macros are rendered from that JSON by a small script (`gen_results_tex.py`), and the difficulty-atlas figure is rendered from the same package’s difficulty and link functions (`gen_figure.py`); `make repro` runs the whole chain and rebuilds the PDF. Because the reported numbers are generated from the code rather than transcribed, they cannot drift from it. The run is deterministic, completes in under a minute, and uses no proprietary data and no network access.

Recovered versus ground-truth values. For the avoidance of doubt: the values in Table 1 are the estimator’s *recovered* outputs (for example, the difficulty minimum recovered at $\hat{x}^* = 16.7$ kn). The value $x^* = 18$ kn is the *ground truth* supplied to the generator. The recovery claim is precisely that the former lands within ± 3 kn of the latter.

References

- Frank B. Baker and Seock-Ho Kim. *Item Response Theory: Parameter Estimation Techniques*. Marcel Dekker, New York, 2 edition, 2004.
- Allan Birnbaum. Some latent trait models and their use in inferring an examinee’s ability. In Frederic M. Lord and Melvin R. Novick, editors, *Statistical Theories of Mental Test Scores*, pages 397–479. Addison-Wesley, Reading, MA, 1968.
- R. Darrell Bock and Murray Aitkin. Marginal maximum likelihood estimation of item parameters: Application of an EM algorithm. *Psychometrika*, 46(4):443–459, 1981. doi: 10.1007/BF02293801.
- Glenn W. Brier. Verification of forecasts expressed in terms of probability. *Monthly Weather Review*, 78(1):1–3, 1950. doi: 10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2.
- Paul De Boeck and Mark Wilson, editors. *Explanatory Item Response Models: A Generalized Linear and Nonlinear Approach*. Statistics for Social and Behavioral Sciences. Springer, New York, NY, 2004. doi: 10.1007/978-1-4757-3990-9.
- C. R. de Freitas, Daniel Scott, and Geoff McBoyle. A second generation climate index for tourism (CIT): Specification and verification. *International Journal of Biometeorology*, 52(5):399–407, 2008. doi: 10.1007/s00484-007-0134-3.
- Susan E. Embretson and Steven P. Reise. *Item Response Theory for Psychologists*. Lawrence Erlbaum Associates, Mahwah, NJ, 2000.
- Gerhard H. Fischer. The linear logistic test model as an instrument in educational research. *Acta Psychologica*, 37(6):359–374, 1973. doi: 10.1016/0001-6918(73)90003-6.
- Gene H. Golub and John H. Welsch. Calculation of Gauss quadrature rules. *Mathematics of Computation*, 23(106):221–230, 1969. doi: 10.1090/S0025-5718-69-99647-1.
- Frederic M. Lord and Melvin R. Novick. *Statistical Theories of Mental Test Scores*. Addison-Wesley, Reading, MA, 1968.
- Darryl I. MacKenzie, James D. Nichols, Gideon B. Lachman, Sam Droege, J. Andrew Royle, and Catherine A. Langtimm. Estimating site occupancy rates when detection probabilities are less than one. *Ecology*, 83(8):2248–2255, 2002. doi: 10.1890/0012-9658(2002)083[2248:ESORWD]2.0.CO;2.
- Zbigniew Mieczkowski. The tourism climatic index: A method of evaluating world climates for tourism. *The Canadian Geographer*, 29(3):220–233, 1985. doi: 10.1111/j.1541-0064.1985.tb00365.x.
- Georg Rasch. *Probabilistic Models for Some Intelligence and Attainment Tests*. Danmarks Paedagogiske Institut, Copenhagen, 1960. Reprinted 1980, University of Chicago Press.
- Daniel Scott, Michelle Rutty, Bas Amelung, and Mantao Tang. An inter-comparison of the holiday climate index (HCI) and the tourism climate index (TCI) in Europe. *Atmosphere*, 7(6):80, 2016. doi: 10.3390/atmos7060080.
- David I. Warton, F. Guillaume Blanchet, Robert B. O’Hara, Otso Ovaskainen, Sara Taskinen, Steven C. Walker, and Francis K. C. Hui. So many variables: Joint modeling in community ecology. *Trends in Ecology & Evolution*, 30(12):766–779, 2015. doi: 10.1016/j.tree.2015.09.007.